

# ONLINE RETAIL SALES ANALYTICS & DATABASE MANAGEMENT SYSTEM

## Project Report

### 1. Introduction

This project develops a normalized SQL database for an online retail business and uses SQL queries to analyze sales transactions. The project is based on a Kaggle online retail transaction dataset. The raw data contains invoice numbers, stock codes, product descriptions, quantities, invoice dates, unit prices, customer IDs, and countries. The database was designed to separate customers, products, orders, order items, categories, and payments so that the data can be stored with reduced redundancy and queried efficiently.

### 2. Abstract

The project follows a practical data-processing workflow: raw retail data was imported into MySQL, checked for data-quality issues, cleaned, and transformed into a relational database structure. The cleaned sample contained 653 valid transaction rows, representing 31 unique customers, 416 unique products, and 44 unique orders. SQL was then used for sales analysis, including revenue analysis, top-product and top-customer analysis, average order value, category analysis, filtering with HAVING, subqueries, common table expressions (CTEs), and window-function ranking. Reusable SQL views were also created for sales, customer, monthly, and product-performance reporting.

### 3. Tools Used

| Tool / Technology | Purpose                                                                 |
|-------------------|-------------------------------------------------------------------------|
| MySQL 8.0         | Database creation, schema design, SQL queries and analysis              |
| MySQL Workbench   | Database implementation, data import and EER/ER diagram                 |
| Kaggle dataset    | Source online retail transaction data                                   |
| SQL               | Cleaning, transformation, joins, aggregation, CTEs, views and analytics |

### 4. Steps Involved in Building the Project

- **Dataset preparation:** The online retail transaction data was obtained from Kaggle and imported into a raw/staging table.
- **Data cleaning:** Records with invalid quantities or prices, missing customer IDs, missing descriptions, and duplicate records were removed. The cleaned dataset contained 653 valid rows.
- **Database design:** The raw transaction data was normalized into Customers, Categories, Products, Orders, Order\_Items, and Payments tables with primary and foreign-key relationships.
- **Data transformation:** Invoice dates stored as text were converted to MySQL DATETIME values. Products were classified into categories using keyword-based rules derived from their descriptions.
- **Payment modeling:** The source dataset did not contain payment details. Payment method/status values were therefore generated as synthetic project data and are documented as such.
- **SQL analytics:** JOINS, GROUP BY, HAVING, subqueries, CTEs, aggregate functions, date functions, and window functions were used to answer business questions.
- **Reporting views:** Reusable views were created for sales summaries, customer sales summaries, monthly sales, and product performance.
- **ER diagram:** MySQL Workbench was used to generate the relational EER diagram showing the relationships between the six core tables.

### 5. Database Structure and Analysis

**Core tables:** Customers, Categories, Products, Orders, Order\_Items, and Payments. The Order\_Items table acts as the bridge between Orders and Products, allowing one order to contain multiple products and a product to appear in multiple orders. The project includes queries for total revenue, revenue by country and category, top products by revenue and quantity, top customers by spending, average order value, repeat customers, products above average price, and category-wise product ranking.

### 6. Conclusion

The project demonstrates how raw online retail transaction data can be transformed into a structured relational database and analyzed using SQL. The final design separates major business entities, applies primary and foreign keys, and supports reusable analytical views. The project also provides practical experience with data cleaning, normalization, joins, aggregation, CTEs, subqueries, and window functions. The resulting database can be extended with additional customer attributes, product categories, real payment information, inventory data, or larger transaction volumes in a future version.

**Note:** The report describes the project dataset and transformations actually used. Category assignments were derived from product descriptions, while payment attributes were synthetically generated because payment information was not present in the source dataset.